Detection of AI-Generated Images in Social Media

Silesian University of Technology - Department of Computer Graphics, Vision and Digital Systems

AUTHORS

- Błażej Jamrozik
- Bartosz Kępa

SUPERVISOR

Anna Daniłowicz, MSc Eng.

AI-Generated Images in Social Media

In recent years, AI image generators have rapidly improved and can now create highly realistic images, flooding social media with synthetic visual content.

AI GENERATORS

Midjourney

Photorealistic art generation

DALL·E

OpenAI image synthesis

Stable Diffusion

Open-source generation

RESULTING PROBLEMS

Misinformation & Fake News

Synthetic events and manipulated visual content spread rapidly online.

Digital Security

AI-generated faces and fake photos undermine trust in online media.

⚠ Automatic systems capable of detecting AI-generated images are becoming increasingly important.

Technologies and Tools

The project was built entirely in Python, leveraging a modern deep learning stack for model development, data processing, and deployment.

CORE LIBRARIES

PyTorch & torchvision

Deep learning framework and vision utilities.

NumPy & Pandas

Numerical computation and data manipulation.

Matplotlib & scikit-learn

Visualisation and evaluation metrics.

DEPLOYMENT & COLLABORATION

Streamlit

Rapid web application framework for model deployment.

GitHub

Version control and collaborative development platform.

Datasets Used in the Project

The project drew from multiple datasets sourced from Kaggle and Hugging Face, enabling robust cross-dataset evaluation and improved model generalisation.

PARVESHI (default dataset on which the model was trained)

https://huggingface.co/datasets/Parveshiiii/AI-vs-Real

LuckyCow

https://huggingface.co/datasets/LuckyCow/Ai_vs_real_web_images

NanoBanana

https://huggingface.co/datasets/julienlucas/midjourney-dalle-sd-nanobananapro-dataset

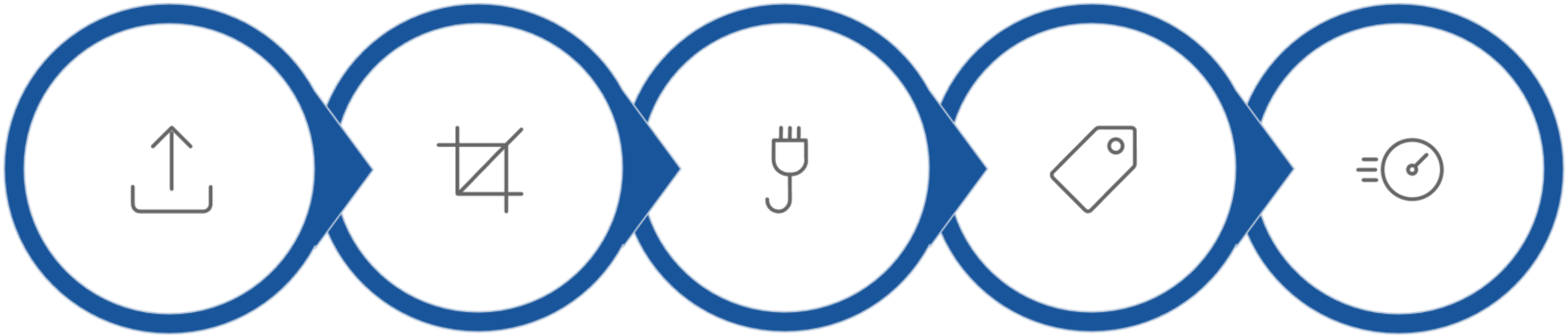
Kaggle 1–3

Kaggle1: https://www.kaggle.com/datasets/tristanzhang32/ai-generated-images-vs-real-images
Kaggle2: https://huggingface.co/datasets/Hemg/AI-Generated-vs-Real-Images-Datasets
Kaggle3: https://www.kaggle.com/datasets/rhythmghai/ai-vs-real-images-dataset

Using multiple datasets allowed cross-dataset evaluation, testing model generalisation, and improving robustness against different AI generation styles.

Application Workflow

The system follows a clear, linear pipeline from image upload through to final prediction and confidence score display.



Upload Image

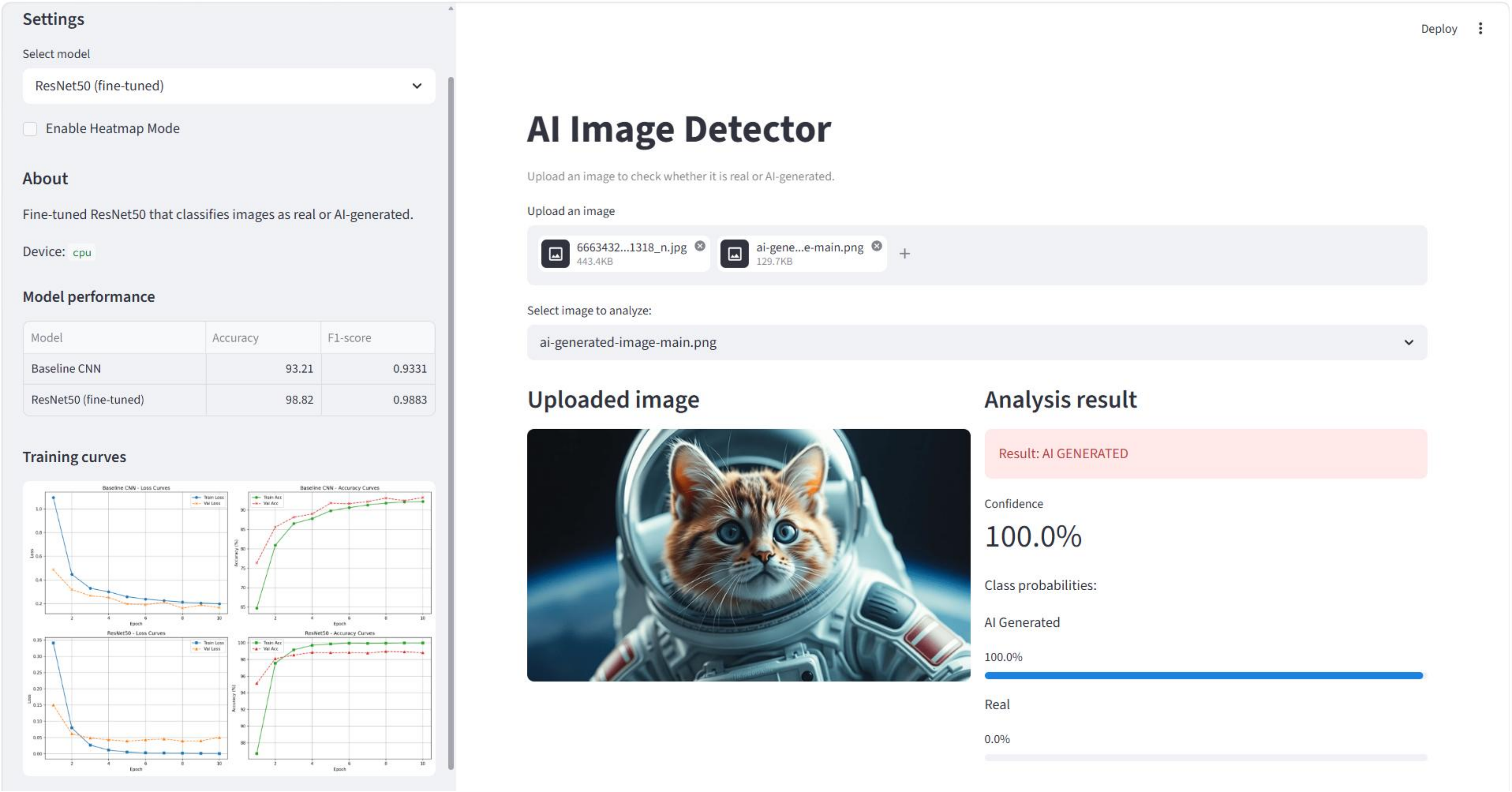
Preprocessing

CNN / ResNet50

Prediction

Confidence Score

Each stage is handled automatically - the user simply uploads an image and receives an instant classification result alongside a confidence percentage, making the tool accessible to non-technical users.


Settings
Select model
ResNet50 (fine-tuned)
Enable Heatmap Mode
About
Fine-tuned ResNet50 that classifies images as real or AI-generated.
Device: cpu
Model performance
Model Accuracy F1-score
Baseline CNN 93.21 0.9331
ResNet50 (fine-tuned) 98.82 0.9883
Training curves
Baseline CNN - Loss Curves
Baseline CNN - Accuracy Curves
ResNet50 - Loss Curves
ResNet50 - Accuracy Curves
Deploy
AI Image Detector
Upload an image to check whether it is real or AI-generated.
Upload an image
6663432...1318_n.jpg
443.4KB
ai-gene...e-main.png
129.7KB
Select image to analyze:
ai-generated-image-main.png
Uploaded image
Analysis result
Result: AI GENERATED
Confidence
100.0%
Class probabilities:
AI Generated
100.0%
Real
0.0%

Data Preprocessing

Before training, all images were preprocessed to ensure consistency and to maximise model performance. Augmentation techniques were applied to increase dataset diversity and reduce overfitting.

Before:

After:

AUGMENTATION TECHNIQUES

Horizontal Flip

Randomly mirrors images to simulate varied perspectives.

Rotation

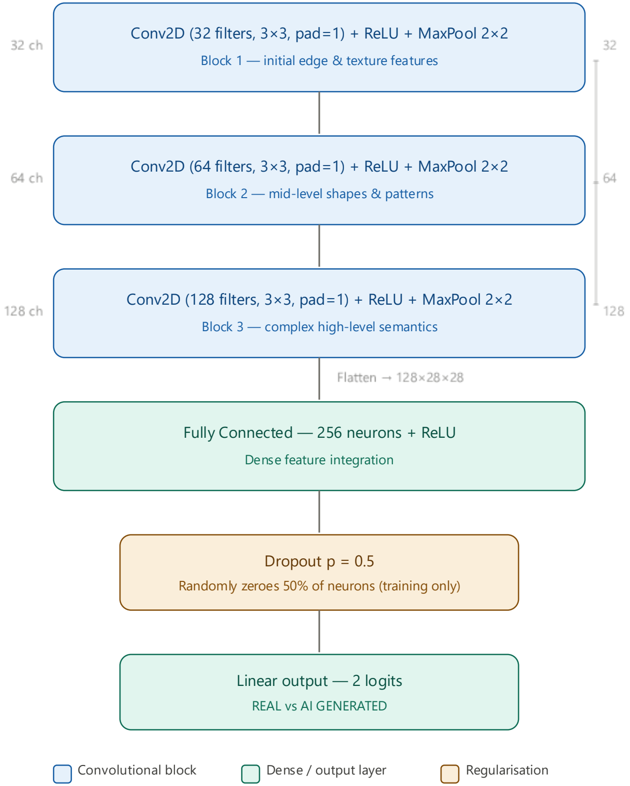
Applies random angular rotations to diversify training samples.

Colour Jitter

Varies brightness, contrast, and saturation for robustness.

Baseline CNN Model

The first implemented model was a simple **Baseline CNN** trained from scratch, serving as a reference point for comparison with more advanced architectures.



ARCHITECTURE DETAILS

01

Conv2D + ReLU + MaxPooling

Initial feature extraction with activation and spatial downsampling.

02

Three Convolutional Blocks

Progressively deeper feature maps capturing complex patterns.

03

Fully Connected Layer

Flattened features mapped to a dense representation.

04

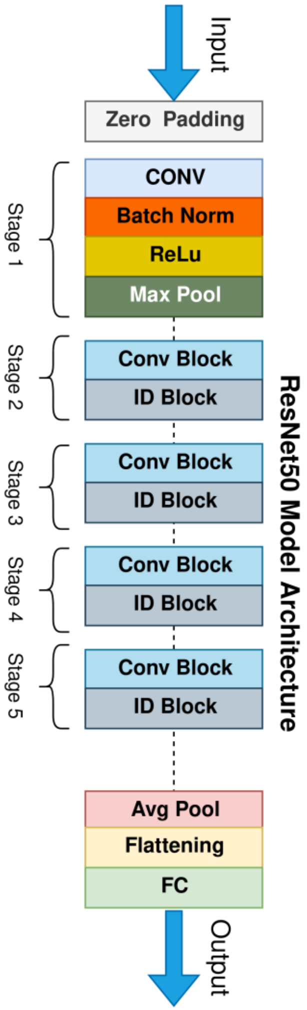
Dropout

Regularisation to prevent overfitting during training.

05

Binary Classification Output

Final sigmoid layer producing REAL vs AI GENERATED probability.



ResNet50 with Transfer Learning

The final solution leveraged ResNet50 with transfer learning, significantly outperforming the Baseline CNN in accuracy and generalisation.

IMPLEMENTATION DETAILS

Pretrained ImageNet Weights

Initialised with rich visual features learned from 1.2 million images.

Frozen Backbone Layers

Early layers kept fixed to preserve low-level feature representations.

Unfrozen layer4 Block

Final residual block fine-tuned on the AI detection task.

Replaced FC Layer

Custom binary classification head replacing the original ImageNet output.

ADVANTAGES OF TRANSFER LEARNING

Faster Training

Pretrained weights dramatically reduce the number of epochs required.

Higher Accuracy

Rich feature representations lead to superior classification performance.

Reduced Hardware Requirements

Less compute needed compared to training a large model from scratch.

Model Comparison: Baseline CNN vs ResNet50

MODEL SELECTION

Compared to the Baseline CNN, **ResNet50** achieved superior results across every measured dimension. Transfer learning significantly improved classification quality.

Higher Accuracy

ResNet50 consistently scored higher accuracy across all test datasets.

Better F1-Score

Improved precision-recall balance, reflecting stronger binary classification.

Greater Stability

ResNet50 maintained consistent performance across varying hyperparameters.

Better Cross-Dataset Performance

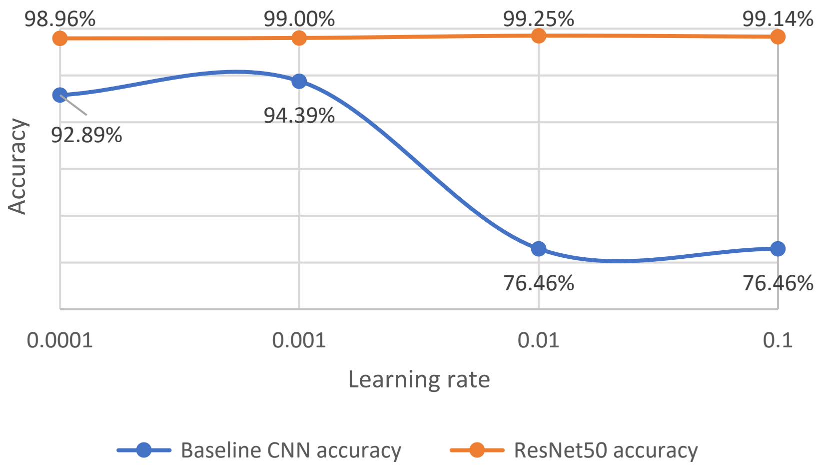
Generalised more effectively to unseen and challenging datasets.

Key Conclusion: Transfer learning significantly improved classification quality - ResNet50 is the clear winner.

Learning Rate Analysis

PARAMETERS AFFECTING PERFORMANCE

Four learning rate values were tested to understand their influence on model performance: 0.0001, 0.001, 0.01, and 0.1.



Key Observations

- Baseline CNN performance **dropped significantly** at high learning rates, while ResNet50 remained relatively stable throughout.
- ResNet50's robustness to learning rate variation is a key advantage of transfer learning - pre-trained weights provide a stable starting point.
- Optimal learning rate range is 0.0001–0.001, especially for simpler architectures like the Baseline CNN.

Baseline CNN Best

Learning rate: 0.001

ResNet50 Default

Learning rate: 0.0001

Image Size Analysis

PARAMETERS AFFECTING PERFORMANCE

Three input image sizes were tested to determine the effect of visual detail on model performance.

1

224 × 224

Standard baseline size. Sufficient for basic feature extraction but misses finer details.

2

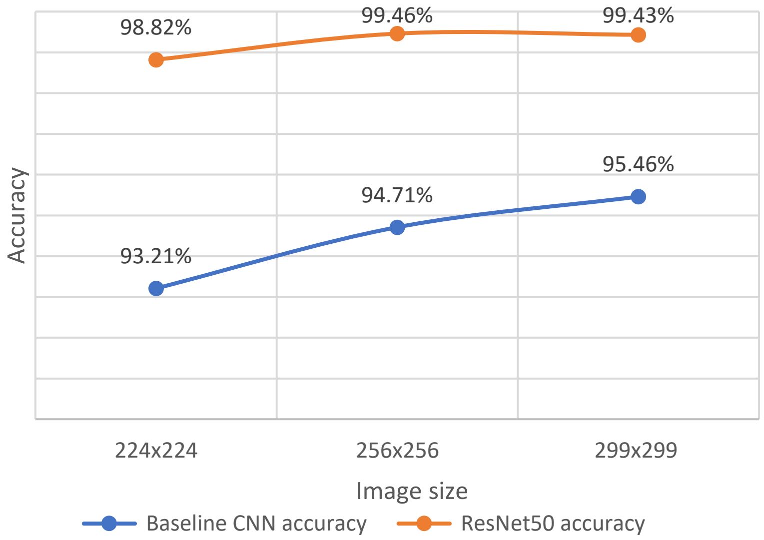
256 × 256

Improved performance. More visual detail available for the network to analyse artefacts.

3

299 × 299

Best results achieved. Higher resolution enables detection of subtle AI-generated image artefacts.

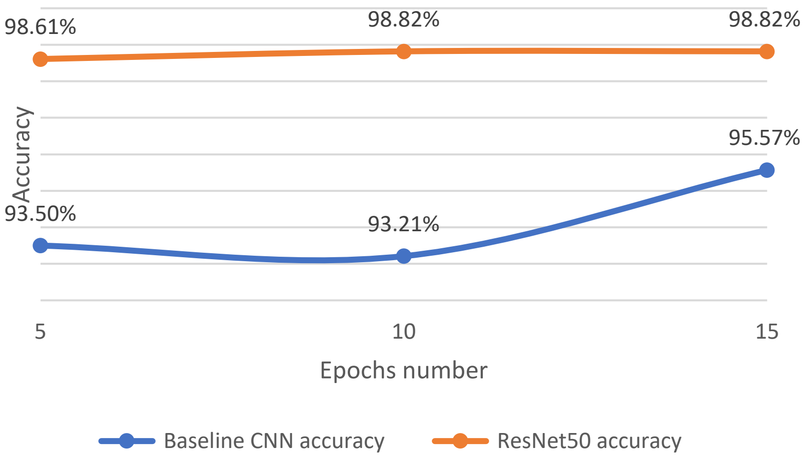


Conclusion: Higher image resolutions improved model performance because the networks could analyse more visual details. Best results were achieved at 256×256 and 299×299.

Training Epoch Analysis

PARAMETERS AFFECTING PERFORMANCE

The number of training epochs was varied to assess convergence speed and final performance for both models.



Observations

ResNet50 - Fast Convergence

Achieved strong performance very quickly, even at just 5 epochs, thanks to pre-trained weights.

Baseline CNN - Gradual Improvement

Improved with additional epochs but still underperformed compared to ResNet50 at every stage.

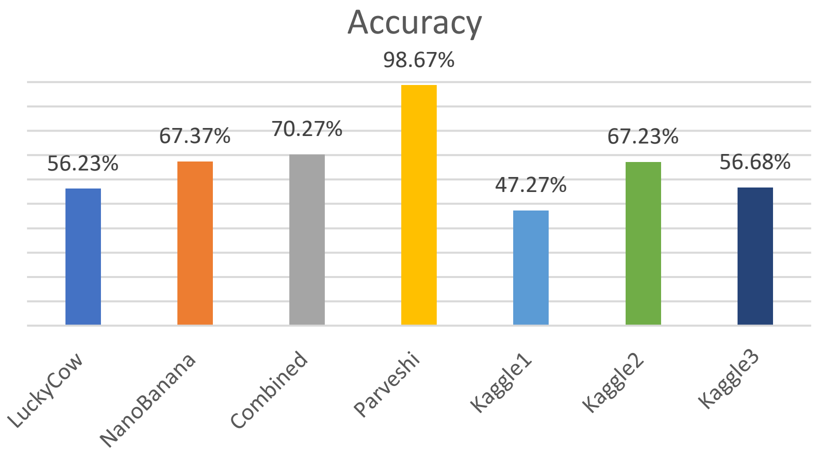
Diminishing Returns

Beyond 10 epochs, gains were marginal - suggesting optimal training efficiency at 10-15 epochs.

Cross-Dataset Results: Generalization Performance

DESCRIPTION

The slide presents model's generalization ability when trained on the **Parveshi dataset** and tested across multiple external datasets. Performance varies significantly depending on the dataset's complexity and similarity to the training data.



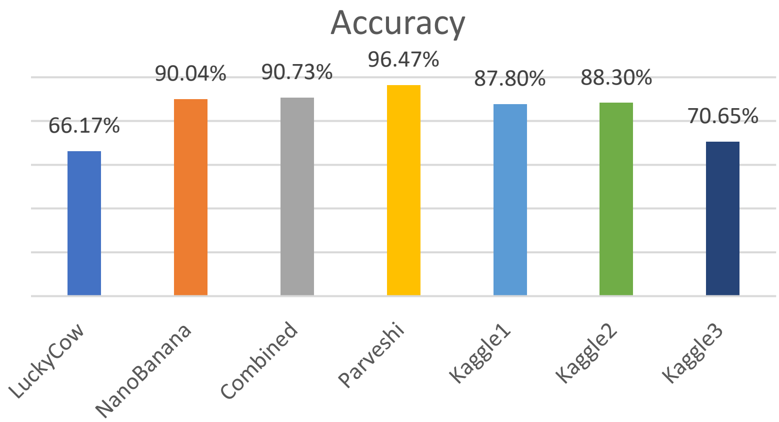
Key Observations

- Best generalization occurred when training on combined datasets, improving robustness across domains.
- Performance drops sharply on datasets with different visual characteristics (e.g. LuckyCow, Kaggle1).
- Resilience to domain shift is moderate - the model retains fair performance on NanoBanana and Kaggle2 (~67%).
- Cross-dataset generalization remains challenging, highlighting the need for more diverse training data or domain adaptation techniques.
- LuckyCow proved the most difficult, confirming that dataset variability strongly affects transfer learning outcomes.

Cross-Dataset Results: Training on Multiple Datasets

DESCRIPTION

This slide presents cross-dataset evaluation results for a model trained on a combined set of images from multiple datasets. Training on diverse data significantly improved the model's ability to generalize across different domains.



Key Observations

- Training on multiple datasets greatly improves generalization, leading to consistently high performance across domains.
- Best results were obtained on NanoBanana and the Combined dataset, indicating that diverse training data enhances feature robustness.
- Performance on the original Parveshi dataset remains high, showing that multi-dataset training does not degrade in-domain accuracy.
- Multi-dataset training is essential for building a model that performs reliably on real-world, heterogeneous data.

Final Conclusions

RESULTS

The research successfully demonstrated the viability of deep learning for AI-generated image detection. The following conclusions were drawn from the experimental results.

Transfer Learning Wins

Transfer learning significantly improves classification performance over training from scratch.

ResNet50 Outperforms

ResNet50 outperformed the custom CNN model across all datasets and metrics.

Generalisation Challenge

Cross-dataset generalisation remains the biggest challenge for robust deployment.

Size Matters

Larger input images improve detection quality by preserving fine-grained artefacts.

Multi-Dataset Robustness

Using multiple datasets during training improves model robustness and generalisation.

The best configuration

LR = 0.001, image size = 256/299 px, epochs = 10-15, multi-dataset training

The system **successfully detects AI-generated images** in most cases, validating the proposed approach.

Thank You for Your Attention

This research demonstrated that deep learning - and transfer learning in particular - provides a powerful and effective approach to detecting AI-generated images across diverse datasets.

7

Datasets Tested

Across varied real-world conditions

Questions? We welcome your thoughts, feedback, and discussion on the findings presented.

Sources

Gamma

The visual layout of the presentation was created with the assistance of Gamma.

Nano Banana Pro

Title slide image was generated with Nano Banana Pro.

Chałkoń

"Challah horse": Chałkoń

Beer Photo

Beer photo: private archive

ResNet50 Model Architecture

The Annotated ResNet-50 (Towards Data Science)

Google Imagen 3

Cyberpunk animals in a ResNet50 vs Baseline CNN duel were generated with Google Imagen 3.

PictoGraphic

The other images are sourced from PictoGraphic.